

Introduction to Machine Learning

Lab 6: Nonlinear Dimensionality Reduction (Kernel PCA, LLE and Laplacian Eigenmaps)

Hongteng Xu

1 Motivation

- Implement three typical manifold learning methods.
- Try to get some feelings about the sensitivity of some methods to the structure of their K-NN graphs.
- Try to overcome the numerical instability of LLE.

2 Tasks

Please read Lecture 7 and 8 carefully before doing this lab work.

1. Implement Kernel PCA with the following kernels:

$$\text{RBF: } K_h(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|_2^2}{h}\right), \quad \text{Linear: } K(\mathbf{x}, \mathbf{x}') = \mathbf{x}^T \mathbf{x}' \quad (1)$$

where h is the hyperparameter.

2. Construct a K-NN graph of the data points.
3. Implement the Locally Linear Embedding (LLE) algorithm.

Step 1 Construct a K-NN graph, for each sample $\mathbf{x}_n \in \mathbb{R}^D$, find its K nearest neighbors $\mathbf{X}_n \in \mathbb{R}^{D \times K}$.

Step 2 Solve the locally linear self-representation problem: for $n = 1, \dots, N$

$$\min_{\mathbf{w}} \|\mathbf{x}_n - \mathbf{X}_n \mathbf{w}\|_2^2, \quad s.t. \quad \sum_{k=1}^K w_k = 1 \quad (2)$$

The solution is $\mathbf{w}_n = \mathbf{C}^{-1} \mathbf{1}_K$, $\mathbf{C} = (\mathbf{X}_n - \mathbf{x}_n \mathbf{1}_K^T)^T (\mathbf{X}_n - \mathbf{x}_n \mathbf{1}_K^T)$

Step 3 Construct the alignment matrix based on $\{\mathbf{w}_n\}_{n=1}^N$ and obtain the latent codes by eigenvalue decomposition.

4. Implement the Laplacian Eigenmaps. The Laplacian matrix can be derived based on the dense similarity matrix or the sparse similarity matrix modulated by the adjacency matrix of the K-NN graph.